

CSE260

Digital Logic Design



SOP & POS

- **Sum-of-Products (SOP) Expression:** a product term or a logical sum (OR) of several product terms.

Examples: $x+yz'$, $xy'+x'yz$, $AB+A'B'$

- **Product-of-Sums (POS) Expression:** a sum term or a logical product (AND) of several sum terms.

Examples: $x(y+z')$, $(x+y')(x'+y+z)$, $(A+B)(A'+B')$

- Every boolean expression can either be expressed as sum-of-products or product-of-sums expression.

Examples:

SOP:	$x'y + xy' + xyz$
POS:	$(x + y')(x' + y)(x' + z')$



MIN & MAX TERM

MIN & MAX TERM

Minterms are sum terms.

For Boolean functions, the minterms of a function are the terms for which the result is 1.

Boolean functions can be expressed as sum- of-Minterms.

Maxterms are Product terms.

For Boolean functions, the maxterms of a function are the terms for which the result is 0.

Boolean functions can be expressed as Products-of-Maxterms.



MIN and MAX

	A	B	C	F
0	0	0	0	1
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	0
7	1	1	1	0

Min Terms : 0,1,4,5
[000,001,100,101] $F = \sum$
(0,1,4,5)

Max Terms : 2,3,6,7
[010,011,110,111] $F = \prod$
(2,3,6,7)

MIN-SOP and MAX-POS

Here, x and y are random variables used for understanding purpose and do not represent any specific function

Minterms				Maxterms	
x	y	term	notation	term	notation
0	0	$x'y'$	m_0	$x+y$	M_0
0	1	$x'y$	m_1	$x+y'$	M_1
1	0	xy'	m_2	$x'+y$	M_2
1	1	xy	m_3	$x'+y'$	M_3

Each minterm is the complement of the corresponding maxterm:

Example: $m_2 = xy'$

$$m_2' = (xy')' = x' + (y')' = x' + y = M_2$$

Conversion between MIN & MAX

$$F2 = \Sigma(m1, m4, m5, m6, m7)$$

- The complement function of F2 is:

$$F2' = \Sigma(m0, m2, m3) = m0 + m2 + m3$$

$$\begin{aligned} F2 &= (m0 + m2 + m3)' \\ &= \mathbf{m0' \cdot m2' \cdot m3'} \\ &= \mathbf{M0 \cdot M2 \cdot M3} \\ &= \Pi(M0, M2, M3) \end{aligned}$$

x	y	z	F2	F2'
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

Every Boolean function can be expressed as either Sum-of-Minterms or Product-of-Maxterms.

MIN-SOP and MAX-POS

	A	B	C	F
0	0	0	0	1
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	0
7	1	1	1	0

Min Terms : 0,1,4,5 [000,001,100,101] $F = \sum (0,1,4,5)$

$$F = A'B'C' + A'B'C + AB'C' + AB'C$$

Max Terms : 2,3,6,7 [010,011,110,111] $F = \prod (2,3,6,7)$

$$F = (A+B'+C)(A+B'+C')(A'+B'+C)(A'+B'+C')$$

Simplified Function to SOP & POS



How to Covert into SOP:

*Check if each term contains all variable, if not then
AND $(x+x')$ if x is the missing term*

Simplified Function, $F=A+B'C$



How To Convert into SOP

- $F = A + B'C$

$$= A(B + B')(C + C') + B'C(A + A')$$

$$= (AB + AB')(C + C') + B'C(A + A')$$

$$= AB(C + C') + AB'(C + C') + B'C(A + A')$$

$$= ABC + ABC' + \textcolor{red}{AB'C} + AB'C' + \textcolor{red}{AB'C} + A'B'C$$

$$= ABC + ABC' + AB'C + AB'C' + A'B'C$$

$$= \sum(1, 4, 5, 6, 7)$$

How to Covert into POS:

1. Often distributive law $(x+yz)=(x+y)(x+z)$ is used
2. If then terms, like x , are missing, OR xx'
3. Each POS is missing a term so OR missing terms, Again apply distributive law

Simplified Function, $F=A+B'C$



How to Convert into POS

$$\begin{aligned}
 & A + B'C \\
 &= (A+B') (A+C) \\
 &= (A+B'+cc') (A+BB'+c) \\
 &= \{(A+B')+c\} \{(A+B')+c'\} \{(A+c)+B\} \{(A+c)+B'\} \\
 &= (A+B'+c) (A+B'+c') (A+c+B) (A+c+B') \\
 &= \begin{pmatrix} A+B'+c \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} A+B'+c' \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} A+B+c \\ 0 & 0 & 0 \end{pmatrix} \\
 &= \pi(2, 3, 0) \\
 &= \pi(0, 2, 3)
 \end{aligned}$$

Another Example

Find SOP and POS for $F(w,x,y,z) = wy + x'z$

SOP

$$\begin{aligned}
 F(w,x,y,z) &= wy + x'z \\
 &= wy[(x+x')(z+z')] + x'z[(w+w')(y+y')] \\
 &= wy[xz + xz' + x'z + x'z'] + x'z[wy + wy' + w'y + w'y'] \\
 &= \overset{1}{w}\overset{1}{x}\overset{1}{y}\overset{1}{z} + \overset{1}{w}\overset{1}{x}\overset{1}{y}\overset{0}{z}' + \overset{1}{w}\overset{0}{x}'\overset{1}{y}\overset{1}{z} + \overset{1}{w}\overset{0}{x}'\overset{1}{y}\overset{0}{z}' + \overset{0}{w}'\overset{1}{x}\overset{1}{y}\overset{1}{z} + \overset{0}{w}'\overset{1}{x}\overset{1}{y}\overset{0}{z}' + \overset{0}{w}'\overset{0}{x}'\overset{1}{y}\overset{1}{z} + \overset{0}{w}'\overset{0}{x}'\overset{1}{y}\overset{0}{z}' \\
 &= wxyz + wxyz' + wx'y z + wx'y z' + w'x'y z + w'x'y z' + w'x'yz + w'x'yz' \\
 &= \sum(15, 14, 11, 10, 9, 3, 1)
 \end{aligned}$$

Another Example

Find SOP and POS for $F(w,x,y,z) = wy + x'z$

POS

• $F(w,x,y,z) = wy + x'z$

$$F(w,x,y,z) = wy + x'z$$

$$= (wy + x')(wy + z)$$

$$= (w+x')(y+x')(w+z)(y+z)$$

$$= (w+x'+yy')(y+x'+ww')(w+z+xx')(y+z+ww')$$

$$= (w+x'+y)(w+x'+y')(y+x'+w)(y+x'+w')$$

$$(w+z+x)(w+z+x')(y+z+w)(y+z+w')$$

$$= (w+x'+y+zz')(w+x'+y'+zz')(y+x'+w+zz')(y+x'+w'+zz')$$

$$(w+z+x+yy')(w+z+x'+yy')(y+z+w+xx')(y+z+w'+xx')$$

$$= (w+x'+y+z)(w+x'+y+z')(w+x'+y'+z)(w+x'+y'+z')$$

$$(w+x'+y+z)(w+x'+y+z')(w'+x'+y+z)(w'+x'+y+z')$$

$$(w+x+y+z)(w+x+y+z')(w+x'+y+z)(w+x'+y+z')$$

$$(w+x+y+z)(w+x'+y+z)(w'+x+y+z)(w'+x'+y+z)$$

$$= \begin{pmatrix} 0 & 1 & 1 & 0 \\ w+x'+y+z \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 1 \\ w+x'+y+z' \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 1 \\ w'+x'+y+z \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 \\ w'+x'+y+z' \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ w+x+y+z \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 0 \\ w+x+y+z' \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ w+x'+y+z \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ w'+x+y+z \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ w'+x'+y+z \end{pmatrix}$$

$$(7, 5, 13, 6, 0, 2, 4, 8, 12)$$